Sirui Zou

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Education

Southwestern University of Finance and Economics **Sep 2024 – Jun 2028 (Expected)**

Bachelor of Economics *Chengdu, China*

- **Academics & Quant**: GPA: 3.9/4.0 | GRE Quant: 170/170 | GMAT FE: 655 (Top 9% Globally, Pre-enrollment)
- **Core Courses**: Macroeconomics (A+), Python (A+), Advanced Math (A), Linear Algebra (A), Probability & Statistics (A-)
- **Scholarships**: Academic Scholarship ('24, '25), Innovation & Entrepreneurship Scholarship ('24)

University of California, Berkeley **Jan 2026 – May 2026**

Visiting Student: BGA Program *Focus: Economics & Computational Social Science* *Berkeley, CA, USA*

- **Core Courses**: Econometrics (A), Computational Models of Cognition (A+), Challenge Lab (A+)

Peking University **Jul 2025 – Aug 2025**

Summer School: Quantitative Finance (coursework at the National School of Development) *Beijing, China*

- **Grade**: **A+** (Top 10/150). Focused on stochastic analysis and quantitative pricing models.

Publications

PersonaForge: Psychology-Grounded Dual-Process Architecture for Personality-Consistent Role-Playing Agents **Findings of ACL 2026**

- Jizhou Tong*, Sirui Zou* **equal contribution* aclanthology.org/2026.findings-acl.386
- **Method**: Designed a three-layer personality architecture (Big Five + Defense Mechanisms + Dynamic State) with a selective dual-process generation mechanism, addressing LLM "persona drift" in long-text role-playing simulations.
- **Results**: Reduced persona drift by **75%** (6.3% vs. 24.8% baseline) across 50-turn dialogues on 88 characters; selective activation achieves **96%** of full-system performance with only **13.4%** token overhead. Validated on RoleBench with **73.2%** pairwise win-rate.

Working Papers

Do Prices Reveal What Insiders Know? A Reexamination of Plott and Sunder (1982) **Working Paper, Oct 2026**

- Sirui Zou siruizou.com | Data and code: github.com/siruizou2005/Plott-Sunder-1982
- **Design**: In this classic laboratory study, prices can separate rational expectations (RE) from prior information (PI) only in downward periods, when insiders' information lowers the asset's value below what uninformed traders would pay. In upward periods, both theories predict the same price, whatever the dividends and priors.
- **Evidence**: Transcribed the original transaction records (987 trades, 61 periods, five markets). In the **17** downward periods behind the original conclusion, prices move from their no-information level toward the RE price in only **one of five** markets. In upward periods, the first trade covers about **30%** of the distance from the prior value to the informed value, and prices close most of the gap during trading.

Research Experience

Generative Market Simulation **Apr 2026 – Aug 2026**

Supervised by Kehang Zhu (Harvard); advised by Prof. John Horton (MIT Sloan)

- **Research Question**: When LLM agents trade in a realistic prediction market with unrestricted action spaces and no hard-coded strategies, which strategies come to dominate (including any beyond known human repertoires), how do prices form, and how does wealth concentrate over time?
- **Method**: Built the testbed, a binary prediction-market engine (collateral-backed YES/NO minting/merging, order-book matching) that LLM agents drive through the same tool interface a human trader would use, with a live dashboard and full replay logs. Core design principle: the researcher designs the environment (information, institutions, incentives) but never scripts agent behavior. Validated that a bare LLM trades rationally before running strategy experiments.

Decoding China's Consumption Policy — Jan 2026 – Present

Project Lead (Advisors: Prof. Hong Zou, Dean of School of Economics, & Dr. Han Xiao, SWUFE)

- **Research Question**: How can China's consumption-promotion policies (which instruments, for which target groups and goods, at what intensity) be measured systematically and reproducibly across large volumes of heterogeneous, often implicitly-worded government documents?
- **Method**: Leading a two-stage text-as-data pipeline on 5M+ government documents (mainly 2006–2024): (1) a fine-tuned BERT classifier, benchmarked against keyword search and LLMs, to screen consumption-promotion policies; (2) LLM-based structured extraction of policy attributes (instrument type, intensity, target population, goods category, intergovernmental citation), with manual gold-standard annotation, cross-model consistency checks, and hallucination controls. Building an independently cross-verified national consumption-policy database for downstream city-panel analysis.

Basic Public Service Equalization and Common Prosperity — Jan 2026 – May 2026

Research Assistant (Advisor: Prof. Wei Fan, SWUFE) · Sichuan Province-Level Research Grant

- **Research Question**: Does inequality in basic public services (BPS) worsen urban-rural income inequality, and which dimensions matter most?
- **Method**: Constructed a city-year panel (280 cities, 2012–2024) with Theil index and BPS Gini coefficients; estimated two-way fixed effects models across six BPS dimensions (education, healthcare, social security, infrastructure, culture, ecology).
- **Conclusion**: BPS inequality significantly exacerbates urban-rural income inequality across all six dimensions, supporting an "inequality → inequality" mechanism; ecological and educational BPS show the largest effects.

Technical & Engineering Projects

ScrollWeaver & Soulverse: Multi-Agent Social Simulation Engine *Team Lead* — **Oct 2025 – Dec 2025**

- **Project Overview**: Designed and developed a multi-agent social simulation engine that breathes life into static texts (novels, worldbuilding documents, lore collections), transforming them into interactive, co-creatable, and exportable living worlds. (**Global 3rd Prize, 2025 Soul AI Agent Competition**)
- **Technical Implementation**: Built a LangChain-based Orchestrator-Performer architecture with FastAPI/WebSockets for real-time streaming and D3.js for dynamic maps. Empowered multi-LLM agents (GPT, Claude) with autonomous logic and social deduction via RAG, ChromaDB/SQLite memory, a 3-layer personality model, and a dual-process cognitive architecture.

Magic Brush: AI-Powered Drawing Tutor for Children *Private Developer* — **Nov 2025 – Dec 2025**

- Competed in the **Doubao × Intel "Make Ideas Real" AI App Creation Challenge**; awarded **Bronze Prize (Top 30)**. Built an AI-powered step-by-step sketch drawing assistant designed to engage children independently through interactive guidance and real-time feedback.

Skills & Interests

Tools: Python (PyTorch, Pandas, NumPy), SQL, Stata, LaTeX, Linux, Docker, Git

Interests: Computational Social Science, Economics, NLP, LLM Agents

Languages: English (IELTS 7.0), Mandarin (Native)